MIT Integration Bee Qualifying Exam Answers **24 January 2023**

1 $\displaystyle\int x^{\frac{1}{\log x}}\,dx = \boxed{ex}$

2 $\displaystyle\int \operatorname{sech}(x)\,dx = \boxed{2\arctan(e^x)}$

3 $\displaystyle\int \frac{e^x}{(1+e^x)\log(1+e^x)}\,dx = \boxed{\log(\log(1+e^x))}$

4 $\displaystyle\int (1+x+x^2+x^3+x^4)(1-x+x^2-x^3+x^4)\,dx = \boxed{x+\frac{x^3}{3}+\frac{x^5}{5}+\frac{x^7}{7}+\frac{x^9}{9}}$

5 $\displaystyle\int_0^4 \binom{x}{5}\,dx = \boxed{0}$

6 $\displaystyle\int (x+\sin(x)+x\cos(x)+\sin(x)\cos(x))\,dx = \boxed{\frac{(x+\sin(x))^2}{2}}$

7 $\displaystyle\int \left(\sin^2(x)+\cos^2(x)+\tan^2(x)+\cot^2(x)+\sec^2(x)+\csc^2(x)\right)\,dx = \boxed{2\tan(x)-2\cot(x)-x}$

8 $\displaystyle\int_0^{2\pi} \lfloor 2023\sin(x)\rfloor\,dx = \boxed{-\pi}$

9 $\displaystyle\int (2\log x+1)e^{(\log x)^2}\,dx = \boxed{xe^{(\log x)^2}}$

10 $\displaystyle\int \left((1-x)^3+(x-x^2)^3+(x^2-1)^3-3(1-x)(x-x^2)(x^2-1)\right)\,dx = \boxed{0}$

11 $\displaystyle\int_{-2023}^{2023} \underbrace{|||||x|-1|-1|\cdots|-1|}_{2023\,(-1)\text{'s}}\,dx = \boxed{2023}$

12 $\displaystyle\int (\sin^6 x + \cos^6 x + 3\sin^2 x \cos^2 x)\,dx = \boxed{x}$

13 $\displaystyle\int (x+e+1)x^e e^x\,dx = \boxed{x^{e+1}e^x}$

14 $\displaystyle\int_0^1 \left(\frac{x^2}{2-x^2} + \sqrt{\frac{2x}{x+1}}\right)\,dx = \boxed{1}$

15 $\displaystyle\int \frac{1+2x^{2022}}{x+x^{2023}}\,dx = \boxed{\frac{1}{2022}\log(x^{2022}+x^{4044})}$

16 $\displaystyle\int (3\sin(20x)\cos(23x) + 20\sin(43x))\,dx = \boxed{\sin(20x)\sin(23x)}$

17 $\displaystyle\int_0^1 \prod_{k=0}^{\infty}\left(\frac{1}{1+x^{2^k}}\right)\,dx = \boxed{\frac{1}{2}}$

18 $\displaystyle\int \frac{\sin x}{2e^x+\cos x+\sin x}\,dx = \boxed{\frac{x-\log(2e^x+\cos x+\sin x)}{2}}$

19 $\displaystyle\int \frac{\log(x/\pi)}{(\log x)^{\log(\pi e)}}\,dx = \boxed{\frac{x}{(\log x)^{\log \pi}}}$

20 $\displaystyle\int_{-3/2}^{-1/2} (x^5+5x^4+10x^3+8x^2+x)\,dx = \boxed{\frac{5}{6}}$